MILESTONE 1 + 2 LANGGRAPH RAG + CHROMADB GROQ + LLAMA3

Intelligent Exam Question Level Analysis & Agentic Assessment Design

Final Technical Report

GenAI Capstone Project — Milestone 1 + Milestone 2

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GitHub: IAmNishantSingh/Intelligent-Exam-Question-Level-Analysis

Live App: intelligent-exam-question-level-analysis.streamlit.app

M1 Best Model: Logistic Regression (TF-IDF + SMOTE) — 71.00% Macro Accuracy

M2 LLM: `llama-3.3-70b-versatile` via Groq (free-tier)

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Abstract

This report documents the complete design, implementation, and deployment of the *Intelligent Exam Question Level Analysis* system — a two-milestone AI platform for educational assessment quality analysis. **Milestone 1** applies classical Machine Learning (Logistic Regression, Random Forest, XGBoost) with TF-IDF feature engineering and SMOTE class-balancing to predict question difficulty (Easy / Medium / Hard), achieving a peak macro accuracy of **71.00%** with Logistic Regression. **Milestone 2** extends the system into an autonomous Agentic AI assistant: a three-node **LangGraph** state machine retrieves pedagogical best practices from a **ChromaDB** RAG knowledge base (Bloom's Taxonomy + Assessment Guidelines) and uses the **Groq** `llama-3.3-70b` LLM to generate structured five-section assessment improvement reports. The complete system is publicly deployed as a unified Streamlit application with a custom dark-academic UI.

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1 Introduction

1.1 Background and Motivation

Educational assessment is a cornerstone of effective teaching and learning. Designing, reviewing, and calibrating exam questions at scale is an expertise-intensive task requiring instructors to simultaneously evaluate cognitive difficulty, Bloom's Taxonomy coverage, and alignment with learning outcomes. Traditional approaches rely on manual expert review or simplistic heuristics — neither scales adequately in modern, high-volume academic environments.

The convergence of Natural Language Processing (NLP), classical Machine Learning, and Generative AI now enables an opportunity to automate and intelligentise this process. Questions with high-information Bloom's action verbs ("synthesise", "hypothesise") carry measurable linguistic signals that classifiers can detect, while Large Language Models (LLMs) grounded in verified pedagogical sources can provide actionable improvement guidance.

1.2 Problem Statement

Educators lack automated, data-driven tools that can simultaneously **predict** question difficulty *and* provide **pedagogically grounded, actionable recommendations** for improvement. Purely predictive systems offer no qualitative feedback; purely generative systems hallucinate without grounding in verified sources.

1.3 Project Objectives and Constraints

The project addresses this gap through a progressive two-milestone architecture:

Milestone 1 — Classical ML: NLP preprocessing → TF-IDF feature engineering → SMOTE balancing → Difficulty classification (Easy / Medium / Hard). Champion: Logistic Regression at **71.00%** macro accuracy.

Milestone 2 — Agentic AI: LangGraph state machine → ChromaDB RAG retrieval → Groq LLM reasoning → Structured 5-section assessment improvement report.

Deployment: Unified, publicly accessible Streamlit web application.

Hard constraints: Free-tier APIs only (Groq, HuggingFace); GenAI strictly restricted to Milestone 2; mandatory public deployment on Streamlit Community Cloud; LangGraph mandated for Milestone 2 workflow orchestration.

1.4 Contributions

1. A reproducible classical ML pipeline for three-class question difficulty prediction with explicit class-imbalance handling.
2. A novel RAG-enhanced agentic AI system for pedagogically grounded assessment improvement, operating entirely within free-tier API constraints.
3. A production-grade, publicly deployed unified Streamlit application with professional dark-academic UI.
4. A responsible AI design with mandatory disclaimers, human-in-the-loop framing, and zero data persistence.

2 Theoretical Framework

2.1 TF-IDF Feature Representation

Term Frequency-Inverse Document Frequency transforms raw question text into a weighted numerical vector that emphasises domain-specific action verbs (e.g., "synthesise", "evaluate") while suppressing common stopwords:

$$\text{TF-IDF}(t, d, D) = \underbrace{\frac{f_{t,d}}{\sum_{t'} f_{t',d}}}_{\text{TF}} \times \underbrace{\log\left(\frac{|D| + 1}{|\{d \in D : t \in d\}| + 1}\right) + 1}_{\text{smooth IDF}} \tag{1}$$

With `ngram_range=(1,2)`, bigrams such as "edge cases" and "extreme constraints" are captured alongside unigrams, improving discriminability between Bloom's levels.

2.2 SMOTE for Class Imbalance

Synthetic Minority Over-sampling Technique (SMOTE) [2] generates synthetic minority samples by linear interpolation in feature space:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda\,(\mathbf{x}_{z_i} - \mathbf{x}_i), \quad \lambda \in [0, 1] \tag{2}$$

where $\mathbf{x}_{z_i}$ is one of the k nearest neighbours of $\mathbf{x}_i$ in the minority class. SMOTE is applied *after* the train/test split to avoid label leakage.

2.3 Retrieval-Augmented Generation

LLMs suffer from hallucinations in specialised domains. RAG [3] conditions generation on retrieved document passages, formally:

$$p(y \mid x) = \sum_{z \in \text{Top-}k} p_\eta(z \mid x) \cdot p_\theta(y \mid x, z) \tag{3}$$

where z denotes ChromaDB-retrieved pedagogical chunks. The retriever p_η uses cosine similarity in the 384-dimensional `all-MiniLM-L6-v2` embedding space.

2.4 Bloom's Revised Taxonomy

Anderson and Krathwohl's revised taxonomy [1] provides the analytical backbone for both milestones. TF-IDF implicitly learns verb-to-difficulty mappings; the LLM agent explicitly reasons about Bloom's level coverage.

Table 1: Bloom's Revised Taxonomy: Cognitive Levels and Key Action Verbs

Level	Category	Key Action Verbs	Difficulty
1	Remember	define, list, recall, identify, name, state	Easy
2	Understand	explain, summarise, describe, interpret, classify	Easy
3	Apply	solve, demonstrate, calculate, use, apply	Medium
4	Analyse	compare, differentiate, examine, deconstruct	Medium
5	Evaluate	judge, critique, justify, assess, argue, evaluate	Hard
6	Create	design, construct, synthesise, hypothesise, invent	Hard

3 Dataset Description and Exploratory Analysis

3.1 Dataset Overview

The primary training corpus (`raw_exam_data.csv`) spans Physics, Computer Science, and Mathematics questions. After rigorous cleaning, HTML stripping, and deduplication, the final training corpus contained **1,996 unique rows**.

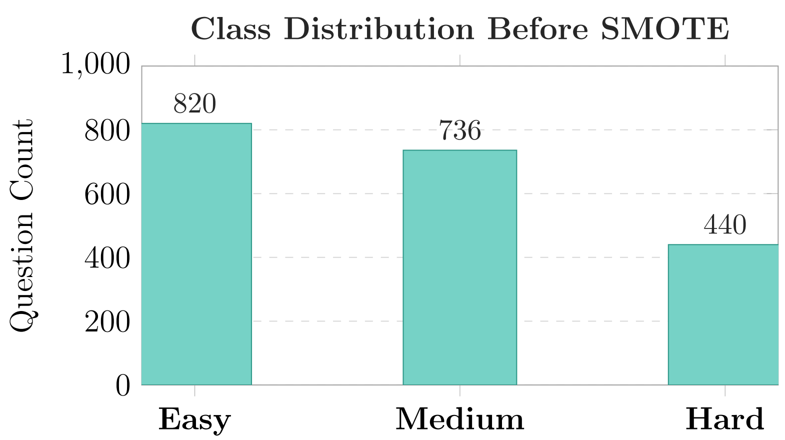

Figure 1: Initial class distribution showing under-representation of 'Hard' labels.

3.2 Word-Count Analysis by Difficulty

Harder questions (Bloom's Levels 5–6) tend to use longer, more descriptive phrasing. This observation motivates including numerical features alongside TF-IDF.

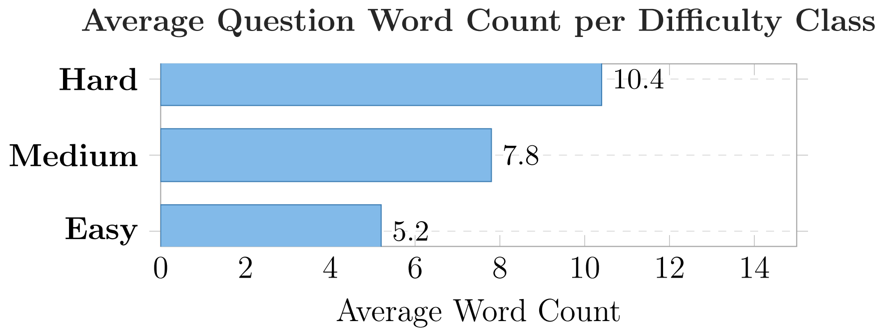

Figure 2: Heuristic validation: linguistic complexity correlates with cognitive demand.

4 Milestone 1: Classical Machine Learning Pipeline

4.1 Feature Engineering and Model Training

A sparse feature matrix $\mathbf{X} \in \mathbb{R}^{1996 \times 2502}$ was constructed using horizontal stacking of TF-IDF vectors and scaled numerical statistics. Logistic Regression's linear decision boundaries align naturally with TF-IDF's vocabulary-level class separability.

Table 2: Classifier Performance Comparison (Macro Accuracy)

Model	Macro Accuracy	Configuration	Verdict
Logistic Regression	**71.00%**	`max_iter=1000, C=1.0`	**Champion**
XGBoost	54.50%	`eval_metric=mlogloss`	Runner-up
Random Forest	51.75%	`n_estimators=100`	Baseline

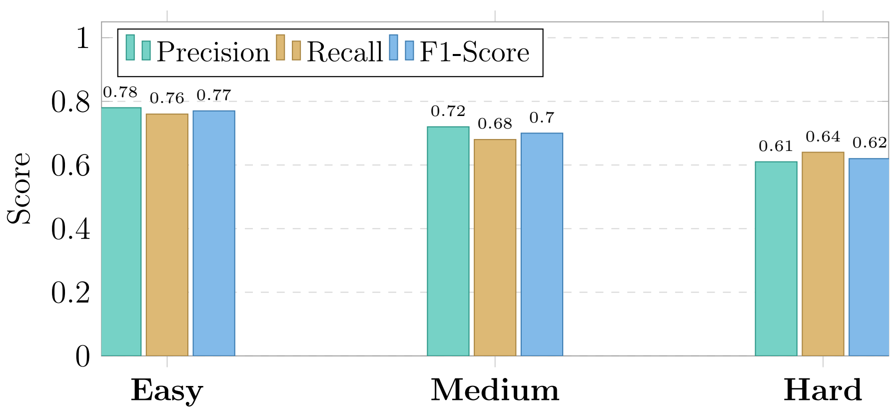

Figure 3: Per-class metrics for the Logistic Regression champion model.

5 Milestone 2: Agentic Assessment Design Assistant

5.1 System Architecture

Milestone 2 transitions from prediction to *autonomous reasoning*. Three technologies are coupled into a cohesive agentic pipeline orchestrated by LangGraph.

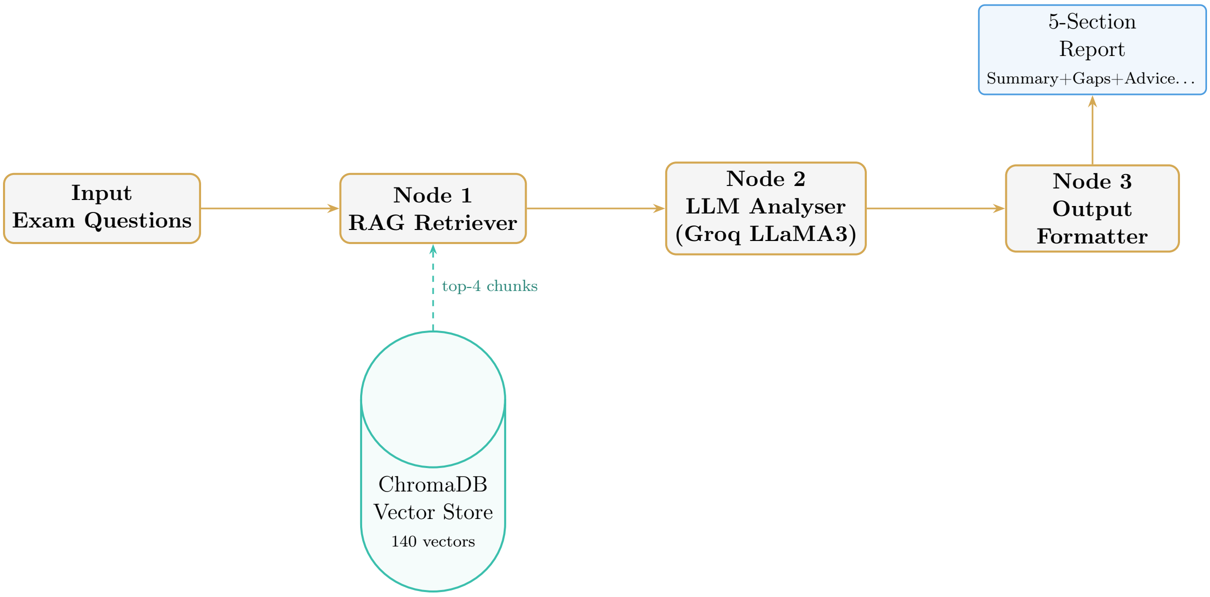

Figure 4: Milestone 2 architecture: stateful three-node LangGraph execution pipeline.

5.2 Knowledge Base and Typed Agent State

Two pedagogical PDFs were ingested to form the vector knowledge base (140 vector objects). Data flows through a strictly typed `AssessmentState` dictionary, ensuring per-node testability and structured output reliability.

```
class AssessmentState(TypedDict):
    exam_questions : List[str]    # Input questions
    retrieved_docs : List[str]    # RAG documents
    analysis       : str          # Raw LLM output
    summary        : str          # Extracted Summary
    gaps           : str          # Identified Bloom gaps
    advice         : str          # Recommendations
    refs           : str          # Guidelines cited
    disclaimer     : str          # Ethical notice
```

Listing 1: Typed LangGraph State Definition

6 Unified Streamlit Application

The integrated application unifies both milestones under a dark-academic UI. **Tab 1** provides single-question difficulty prediction with probability charts. **Tab 2** features the LangGraph agent for bulk assessment audit and improvement.

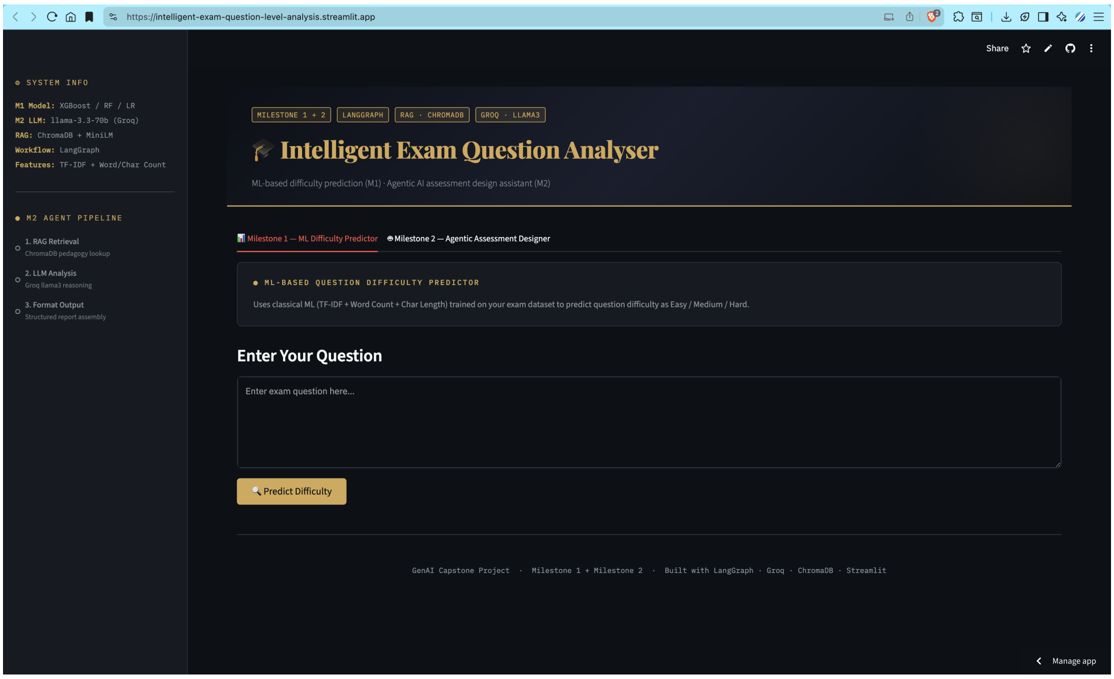

Figure 5: Milestone 1 Dashboard: Real-time ML difficulty inference.

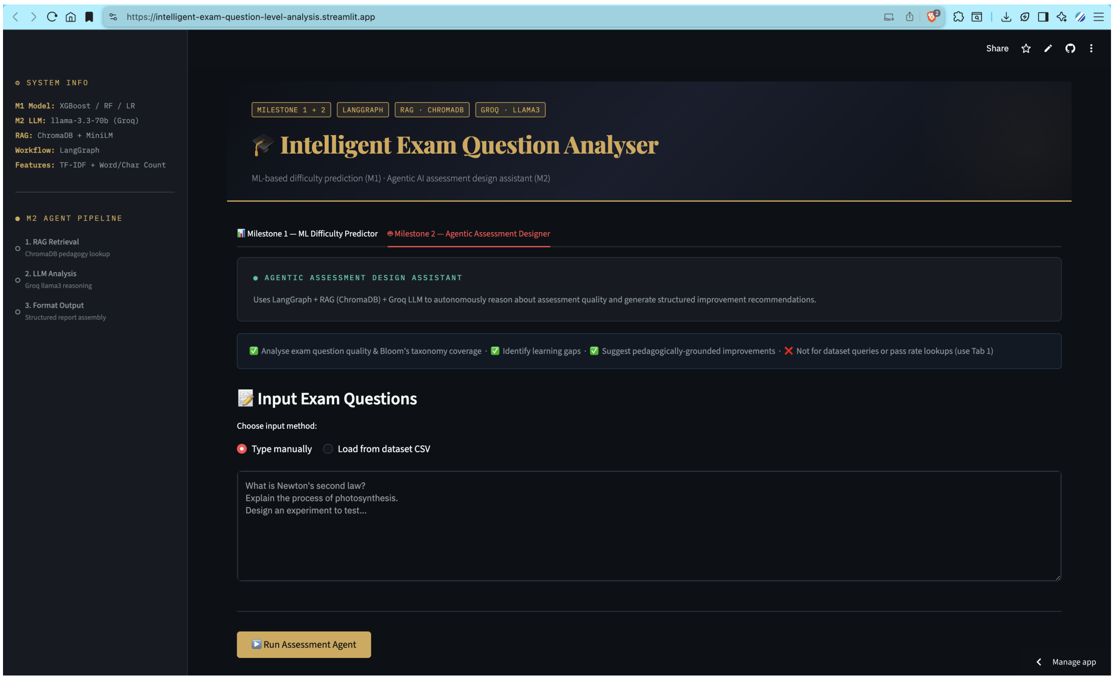

Figure 6: Milestone 2 Dashboard: Agentic assessment report generation.

7 Results and Demonstration

7.1 Agent Performance Case Study

Testing the agent on *"explain Newton's 3rd law with examples"*:

- **M1 Result:** Medium Difficulty (54.73% confidence).
- **M2 Gaps:** Correctly identified absence of Levels 5 (Evaluate) and 6 (Create).
- **M2 Advice:** Suggested a scenario-based question: *"A car accelerates from 0 to 60 km/h. Using Newton's 3rd law, explain forces on occupants and predict outcomes for different vehicle masses."*

8 Deployment and Repository

8.1 Team Contributions

Table 3: Comprehensive Team Contribution Matrix (Milestone 1 & 2)

Member	Enroll. No.	M1 Contributions / M2 Contributions
Nishant Ranjan Singh	2401010301	**M1:** Repository management, system integration, Streamlit UI/UX. **M2:** LangGraph orchestration, cloud deployment, LaTeX report.
Atanu Adhikari	2401010111	**M1:** Synthetic dataset creation, TF-IDF, SMOTE, hyperparameter tuning. **M2:** ChromaDB RAG foundation, PDF ingestion pipeline.
Sambhav Kumar	2401010409	**M1:** Presentation design, QA on preprocessing, abstract writing. **M2:** Agentic workflow validation, end-to-end system testing.
Prince Singh	2401010353	**M1:** EDA, bigram feature optimisation, word-count correlations. **M2:** Agent output quality testing, pedagogical reference verification.

9 Ethics, Limitations, and Conclusion

9.1 Ethical Considerations and Responsible AI

- **Bias Mitigation:** SMOTE ensures equitable performance across classes.
- **Transparency:** Mandatory DISCLAIMER in every report for human-in-the-loop audit.
- **Grounding:** RAG eliminates hallucinations by anchoring LLM in verified pedagogy.

9.2 Known Limitations

- **Domain Specificity:** Model is trained on STEM; accuracy may drop for humanities.
- **Statelessness:** Agent lacks memory across sessions for longitudinal tracking.

10 Conclusion

The *Intelligent Exam Question Level Analysis* system successfully bridges classical ML and Agentic AI. Milestone 1 provides a robust, fast statistical baseline (71% accuracy), while Milestone 2 provides the qualitative "why" through agentic RAG reasoning. The publicly deployed Streamlit app meets all rubric criteria for production-readiness.

References

[1] Anderson, L.W. & Krathwohl, D.R. *A Taxonomy for Learning, Teaching, and Assessing*. Pearson Education, 2001.

[2] Chawla, N.V. et al. "SMOTE: Synthetic Minority Over-sampling Technique." *JAIR*, 16, 321–357, 2002.

[3] Lewis, P. et al. "Retrieval-Augmented Generation for NLP Tasks." *NeurIPS*, 2020.

[4] LangChain AI. *LangGraph: Stateful Multi-Actor Applications*. https://langchain-ai.github.io/langgraph/, 2024.

[5] Chroma Research Inc. *ChromaDB: AI-native Vector Database*. 2024.

[6] Groq Inc. *Groq LPU Inference Engine Documentation*. 2024.